

Password Strength Classification Using Structural Features: A Comparative Study of Machine Learning and Stacking Ensemble Models

Md. Saiful Islam
Computer Science and Engineering
East West University
Dhaka, Bangladesh
2023-3-60-043@std.ewubd.edu

Github: <https://github.com/mohammednaemog-sys>

Rakib Hossen
Asst. Professor
Computer Science and Engineering
East West University
Dhaka, Bangladesh
rakib0001@bdu.ac.bd

Abstract—To keep our accounts and personal information secure, one of the primary ways is a password. The main objective of our study is to see how well machine learning models can determine the strength of a password using its structural features, such as length, character type, etc. For that, we use two publicly available datasets from Kaggle—*Leaked Passwords Largest Datasets*[1] and *Ultimate Cybersecurity Password and Username Dataset*[2]. After data cleaning and removing duplicates, we obtain the dataset containing 3,590,654 unique passwords.

For each password, eight important features were extracted. These include the password length, digit count, lowercase count, uppercase count, special-character count, repeated characters count, maximum consecutive characters, and character-type transitions. After that, we use zxcvbn to generate password strength labels and then categorize them as weak or strong. For training and evaluating, we use six machine learning models such as Logistic Regression, Linear Support Vector Machine, Decision Tree, Random Forest, Neural Network, and Stacking Ensemble.

The results show that among all the models, the Stacking Ensemble performed best overall. It achieved 87.78% Accuracy, 80.73% F1-score, 95.95% Recall, and 0.9564 ROC-AUC. The study also evaluated comparative analysis of different models, classification error analysis, feature importance, and computational evaluation. The analysis showed that among the features used, password length played the most important role.

In summary, the study shows that it is possible to successfully classify the strength of passwords determined by zxcvbn using the structural features of passwords. However, an important point here is that these results should not be seen as a direct measure of how easily a password can be hacked or cracked in real life. Rather, it shows how well machine learning models can work in coordination with zxcvbn's labeling method.

Index Terms—Password Security, Password Strength Classification, Machine Learning, Stacking Ensemble, zxcvbn, Feature Engineering

I. INTRODUCTION

One of the most widely used authentication methods for protecting digital systems and sensitive information is passwords. Weak and easy-to-guess passwords are a major problem for cybersecurity. Using such passwords increases the risk of attacks such as password guessing, dictionary attacks, credential stuffing, brute-force attacks, etc.

How secure a password is depends on some of its important features, such as the length of the password, whether it contains upper and lower case letters, numbers or special characters, whether the same character is repeated, and how complex the password is. In general, the longer and more diverse a password is, the more powerful it is. Machine learning has an effective solution to this problem. It can analyze the various characteristics of the password and learn the relationship of the predefined strength category with them. As a result, the model can predict whether a new password is weak or strong. Previous studies have tested the effectiveness of various machine learning models using the structural properties of passwords. These include Support Vector Machine, Logistic Regression, Neural Network, Decision Tree, Random Forest and Stacking Ensemble. The purpose of these studies was to determine which model could most accurately identify the strength of passwords. [3].

However, how well a machine learning model performs in determining password strength depends largely on how labels are created for passwords. That is, if a model achieves very high accuracy, it does not always mean that the model can accurately understand how easily a password can be hacked in real life. Rather, it only shows in many cases that the model has learned the rules of the labeling method used well.

In our study, we examined how effectively different machine learning models can classify passwords as Weak or Strong using the structural properties of passwords. For this purpose, two open datasets from Kaggle are collected and combined. After data cleaning and duplicate removal, a total of 3,590,654 unique passwords were identified. Each password has 8 structural attributes, such as password length, number of digits, number of lowercase and uppercase letters, number of special characters, number of repeated characters, number of most frequent characters, and number of character types. Then zxcvbn is used to determine the strength of passwords, and its score is converted into two classes: Weak and Strong. Then, Logistic Regression, Linear Support Vector Machine, Decision Tree, Random Forest, Neural Network and Stacking

Ensemble, these six machine learning models were trained and evaluated. To measure model performance, we use Accuracy, Precision, Recall, F1-score, ROC-AUC, and Log Loss. The study also analyzed statistical comparison, classification error analysis, feature importance and computational efficiency.

The main objective of this study is to measure how well the structural properties of passwords are helpful in automatically identifying the classes defined by zxcvbn and to compare the performance of different machine learning models. In short, the study sought to find out which model could tell if a password is weak or strong by looking at the characteristics of the password according to zxcvbn's criteria.

II. LITERATURE REVIEW

Recent research has employed a range of machine learning methods and password-strength meters to identify and evaluate the power of passwords. These studies usually use large-scale leaked password datasets to analyze the password patterns of real users. Researchers have used various preprocessing techniques to prepare data, feature engineering methods to derive password properties, and various labeling methods to characterize passwords as weak or strong. Through these different methods, they have tested which techniques and models can most accurately assess the strength of passwords.

In a study on password strength detection [3] analyzed six representative datasets containing 62,440,436 publicly leaked user records. Preprocessing included duplicate removal, password-length-based outlier filtering, and the removal of illegal or non-printable characters. Six machine learning models, including SVM, Logistic Regression, Neural Networks, Decision Trees, Random Forests, and stacking models, were evaluated. After feature extraction, labeling, and under-sampling, Decision Tree and stacking models achieved the best performance, reporting 99.98% accuracy, 100% recall, and 99.76% F1-score. However, the reliability of these results may depend on the underlying password-labeling methodology.

In another research, researchers used the 000webhost leaked password dataset containing 669,639 passwords labeled as weak, medium, or strong [4]. Duplicate and null values were removed, while special characters were retained during preprocessing. SMOTE was applied only to the training data to address class imbalance and prevent data leakage. Random Forest, SVM, CNN, and Logistic Regression were evaluated using features such as leetSpeak-normalized Shannon entropy. Random Forest achieved the highest accuracy of 99.12% [4]. However, the study noted that models may reproduce the original labeling rules rather than accurately measure real-world password predictability.

On an investigation on the limitations of password-strength meters using publicly available breach datasets, including 178, RockYou, XATO, and Cit0day [5]. Non-ASCII and unusually long passwords were removed during preprocessing, while the 4iQ and Collection#1 datasets were used to analyze passwords associated with unique email accounts. The study highlighted that password-strength meters may not accurately represent

actual account security or password guessability, emphasizing the importance of realistic attack-based evaluation.

Some studies examined password-strength assessment for English, Chinese, and multilingual passphrases. More than 2.4 million unique tokens from English and Chinese FrequencyWords dictionaries were used to generate four-word passphrases with approximately 80 bits of entropy [6]. The study evaluated zxcvbn and MultiPSM and found that password-strength meters may perform differently across languages. zxcvbn performed well for English but less effectively for Chinese, while MultiPSM performed strongly in both guessing scenarios but encountered errors with some Chinese inputs. The study also found that approximately half of the tested websites accepted multilingual passphrases.

So, those studies show that machine learning can achieve higher performance in the case of password strength prediction. But regarding predefined labeling, real-world password predictability, dataset generalization, and multilingual password evaluation, the challenges remain. These limitations indicate the need for more well-approached solutions, such as diverse datasets, comprehensive features, etc.

III. DATASET DESCRIPTION

Two publicly available Kaggle datasets were used in this study. The first dataset, *Leaked Passwords Largest Datasets*[1], consists of two CSV files: `600000-password-data.csv`, containing 669,879 rows and 7 columns, and `10000_common_passwords.csv`, containing 10,000 rows and 1 column. Thus, this dataset initially contained 679,879 password records. During data inspection, one missing password record and six columns containing NaN values were identified in the `600000-password-data.csv` file.

The second dataset, *Ultimate Cybersecurity Password and Username Dataset*[2], consists of two TXT files: `2151220-passwords.txt`, containing 2,151,220 rows with one logical field, and `1000000-password-seclists.txt`, containing 1,000,000 rows with one logical field. Therefore, the second dataset contained 3,151,220 password records, with no missing password records identified.

IV. METHODOLOGY

A. Data Preprocessing

The datasets were preprocessed to improve data quality and ensure the reliability of the subsequent analysis. At first, missing and empty password entries were identified. There is just one missing password, and zero empty password records were found and removed. Duplicate passwords were also identified and removed from a total of 828 duplicate records.

Next, for control and non-printable characters, the datasets were examined. Records containing such characters were removed so that password data contained only valid and usable entries. After cleaning the individual datasets, the processed datasets were combined and duplicate records were removed.

After completing all preprocessing steps, the final dataset contained 3,590,654 unique password entries, with no missing values or duplicate records.

B. Feature Engineering

Feature engineering was performed to transform the cleaned password dataset into a numerical representation which is suitable for machine learning models. Since this study addresses a binary classification problem, each password was assigned one of two target labels: *Weak* or *Strong*.

1) *Password Strength Labeling*: Password strength was estimated using the `zxcvbn` password strength estimation method, which assigns a score ranging from 0 to 4 based on various characteristics of a password, including its length, character composition, common patterns, and estimated resistance to guessing attacks. To express the issue as a binary classification problem, the original scores were grouped into two classes. Passwords with scores from 0 to 2 were categorized as *Weak* and assigned the label 0, whereas passwords with scores of 3 or 4 were categorized as *Strong* and assigned the label 1.

The resulting class distribution is shown below:

TABLE I
DISTRIBUTION OF PASSWORD STRENGTH CLASSES

Password Class	Number of Passwords	Percentage
Weak (Label 0)	2,632,494	73.75%
Strong (Label 1)	958,160	26.25%
Total	3,590,654	100%

2) *Structural Feature Extraction*: Following the password strength labeling process, each password string was transformed into a set of numerical characteristics sample that could be used as input features for the machine learning models. A total of eight structural features were extracted from each password. These features capture different aspects of password composition and structural complexity.

The extracted features are described as follows:

- 1) *Password Length*: Represents the total number of characters in the password.
- 2) *Digit Count*: Represents the total number of numeric digits contained in the password.
- 3) *Lowercase Count*: Represents the number of lowercase alphabetic characters.
- 4) *Uppercase Count*: Represents the number of uppercase alphabetic characters.
- 5) *Special Character Count*: Represents the number of special characters or symbols contained in the password.
- 6) *Repeated Character Count*: Represents the number of repeated characters occurring within the password.
- 7) *Maximum Consecutive Identical Characters*: Represents the length of the longest consecutive sequence of identical characters within the password.
- 8) *Character-Type Transitions*: Represents the number of transitions between different character types, such as lowercase letters, uppercase letters, digits, and special characters.

In this study feature matrix contained 3,590,654 samples and 8 numerical features after feature extraction. The final feature matrix was analyzed to identify data quality with no missing values. This structured numerical representation was subsequently used as the input for training and evaluating the machine learning models.

C. Training, Validation, and Test Data Split

For a reliable model development and evaluation, the complete dataset was split into three subsets those are train, validation, and test. A *stratified splitting strategy* was employed to preserve the original distribution of the two password strength classes, as *Weak* and *Strong*, through our all subsets.

In the training set there was 2,513,457 samples, depicting 70% of the complete dataset, and was used for training the machine learning models. The validation set has 538,598 samples, depicting 15% of the dataset, and was used for model comparison, hyperparameter tuning, and model selection. Finally, the test set has 538,599 samples, also depicting roughly 15% of the complete dataset, and was reserved for the final evaluation of the selected models.

Table II presents the distribution of samples across the three subsets.

TABLE II
TRAINING, VALIDATION, AND TEST DATA DISTRIBUTION

Dataset Subset	Number of Samples	Percentage
Training Set	2,513,457	70%
Validation Set	538,598	15%
Test Set	538,599	15%
Total	3,590,654	100%

The training set was used in order to learn the underlying patterns connected with password strength, the validation set was used to compare model performance and for selecting a suitable model configurations. The test set was not exposed during the training and validation phases and was used only for the final performance evaluation of the selected machine learning models. This division ensures an impartial estimation of each model's capacity to adapt to new password samples that they haven't encountered before.

D. Feature Scaling

Feature scaling was performed to standardize the eight engineered password features before model training. Since the extracted features have different numerical ranges and distributions, standardization was necessary to place them on a comparable scale and prevent features with larger numerical values from affecting influencing the learning process.

To standardize each feature the `StandardScaler` was used by the following equation:

$$z = \frac{x - \mu}{\sigma} \quad (1)$$

where x represents the original feature value, μ denotes the mean of the feature, σ represents the standard deviation, and z is the resulting standardized value.

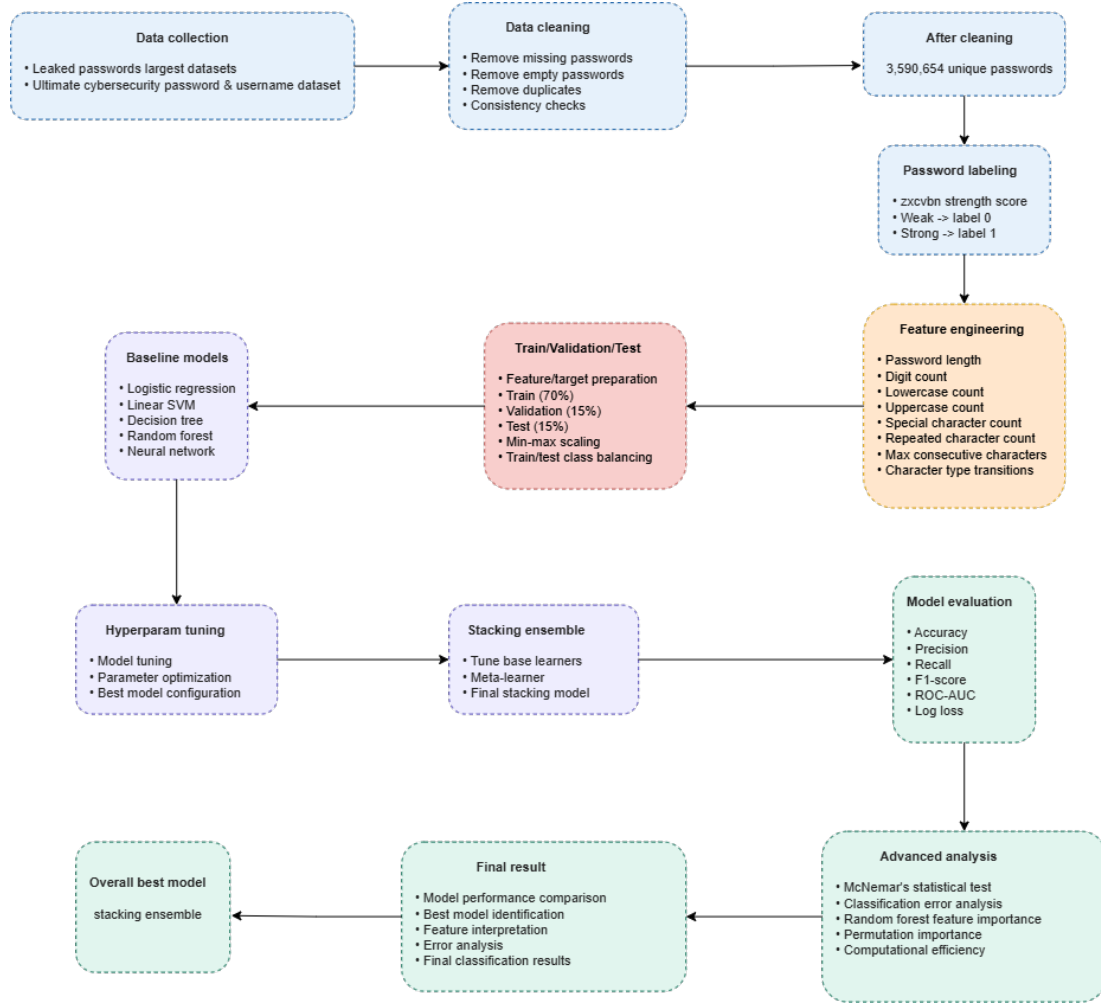


Fig. 1. Overall Methodological Pipeline of the Proposed Password Strength Classification Framework, Illustrating Dataset Preparation, Password Labeling, Structural Feature Engineering, Machine Learning Model Training, Stacking Ensemble Development, Performance Evaluation, and Advanced Model Analysis.

To avoid the leakage of information, the scaling parameters were calculated exclusively from the training dataset. In particular, the mean and standard deviation for each feature were fitted using only the training samples. The same learned transformation parameters were then processed using the previously calculated statistics, without recalculating them for the validation and test datasets.

The training features got around zero mean and unit standard deviation. The scaling process modified only the numerical values of the features and did not alter the number of samples or the dimensionality of the feature matrix. Therefore, the standardized dataset retained 3,590,654 samples and 8 features, which were subsequently used as input for the machine learning models.

E. Dataset Balancing

The training dataset exhibited a noticeable class imbalance, with Weak passwords representing 73.32% of the training samples and Strong passwords representing 26.68%. Such an imbalance which can cause the machine learning models to

become overly influenced by the majority Weak class, and reducing their ability to effectively identify Strong passwords.

To tackle this issue, random undersampling was applied to the training dataset. The majority class, consisting of Weak passwords, was randomly reduced to match the number of samples in the minority Strong password class. The validation and test datasets did not get undersampling and retained their original class distributions to ensure that model comparison. The final evaluation were performed on data reflecting the natural distribution of password strength classes.

After the undersampling process, both classes contained 670,712 samples, resulting in a balanced training dataset with an equal class distribution of 50.00% for each class. The final balanced training dataset contained a total of 1,341,424 samples.

Following dataset balancing, the resulting training feature matrix had a shape of (1,341,424, 8), indicating 1,341,424 training samples and 8 engineered input features. This balanced dataset was subsequently used for training the machine learning models.

TABLE III
CLASS DISTRIBUTION AFTER RANDOM UNDERSAMPLING

Class	Number of Samples	Percentage
Weak (Label 0)	670,712	50.00%
Strong (Label 1)	670,712	50.00%
Total	1,341,424	100.00%

F. Baseline Model Training and Evaluation

Six machine learning models were trained using the balanced training dataset consisting of 1,341,424 samples. The selected models represent different machine learning approaches, including linear, tree-based, neural network, and ensemble-based methods. The models used in this study were Logistic Regression, Linear Support Vector Machine (SVM), Decision Tree, Random Forest, Neural Network, and a Stacking Ensemble.

Model were trained using the same set of eight engineered and standardized password features to ensure a fair comparison. Model performances were subsequently evaluated using the validation dataset. The validation results were used to compare the performance of the models and support the selection of the most suitable approach.

Six evaluation metrics were used: Accuracy, Precision, Recall, F1-score, ROC-AUC, and Log Loss.

Accuracy was measured and the overall proportion of correctly classified samples and is defined as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

Precision evaluates the ratio of predicted positive instances that were correctly classified:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3)$$

Recall evaluates the ratio of predicted positive instances that were correctly identified:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4)$$

The F1-score signifies the harmonic average of Precision and Recall:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (5)$$

The ROC-AUC measures the area under the Receiver Operating Characteristic (ROC) curve and evaluates model's ability to find difference between the two classes across different classification thresholds.

Log Loss evaluated the quality of the predicted probabilities by penalizing predictions that assign low probabilities to the correct class. It is calculated as:

$$\text{Log Loss} = -\frac{1}{N} \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)] \quad (6)$$

where TP signifies True Positives, TN signifies True Negatives, FP signifies False Positives, and FN signifies False Negatives. In the Log Loss equation, y_i represents the actual class label of sample i , p_i represents the probability of the prediction for that particular sample, and N represents the total number of evaluated samples.

Using multiple evaluation metrics allowed the models to be compared from different perspectives, including overall classification performance, the ability to correctly identify Strong passwords, the balance between Precision and Recall, discrimination capability, and the quality of predicted probabilities.

V. RESULTS ANALYSIS

The six baseline machine learning models were evaluated on the validation dataset using Accuracy, Precision, Recall, F1-score, ROC-AUC, and Log Loss. The results are presented in Table IV.

TABLE IV
PERFORMANCE COMPARISON OF BASELINE MODELS ON THE VALIDATION SET

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC	Log Loss
Logistic Regression	86.25%	69.51%	86.36%	77.03%	0.9364	0.365950
Linear SVM	86.24%	69.57%	86.10%	76.96%	0.9364	0.365625
Decision Tree	86.87%	67.45%	98.16%	79.96%	0.9540	0.355020
Random Forest	86.88%	67.41%	98.40%	80.01%	0.9552	0.291938
Neural Network	86.59%	66.84%	98.72%	79.71%	0.9538	0.258147
Stacking Ensemble	87.65%	69.45%	95.90%	80.56%	0.9555	0.260960

After considering all results, the Stacking Ensemble achieved the highest validation Accuracy of 87.65% and the highest F1 Score of 80.56%. It also achieved the highest ROC-AUC value of 0.9555, indicating strong discriminative capability between Weak and Strong passwords. Neural Network achieved the highest Recall of 98.72%, which demonstrates its strong ability to identify Strong passwords. Moreover, it produced a very low Log Loss value of 0.258147, indicating a better probabilistic prediction performance among the evaluated baseline models. Random Forest got a competitive performance, with an Accuracy of 86.88%, an F1-score of 80.01%, and a ROC-AUC value of 0.9552. The Decision Tree also demonstrated a high Recall of 98.16%, although its Precision was comparatively lower.

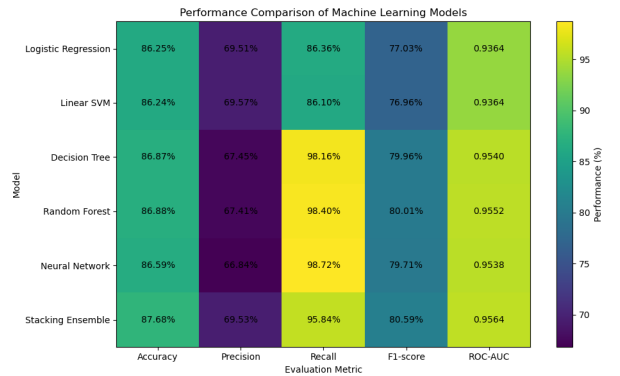


Fig. 2. All model performance comparison.

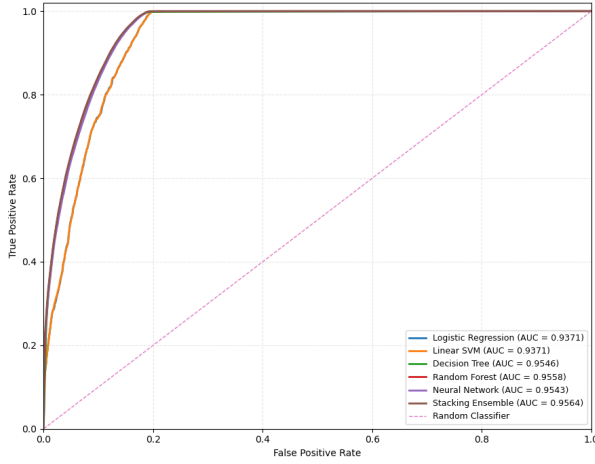


Fig. 3. ROC Curve comparison.

Overall, the baseline results showed that the ensemble based and neural network models provided the strongest overall performance. These baseline evaluation results were subsequently used to guide the hyperparameter tuning process and final model selection. Log Loss was included as an important evaluation metric to assess the quality of the probability generated by the six models. Unlike classification metrics that evaluated predictions based on a specific decision threshold, Log Loss evaluates how well the predicted probabilities correspond to the actual class labels. Lower Log Loss values indicate better probabilistic prediction performance. The Log Loss results for the baseline models are presented in Table V.

TABLE V
LOG LOSS COMPARISON OF BASELINE MODELS

Model	Log Loss
Neural Network	0.258147
Stacking Ensemble	0.260960
Random Forest	0.291938
Decision Tree	0.355020
Linear SVM	0.365625
Logistic Regression	0.365950

Neural Network achieved the lowest Log Loss value of 0.258147 among all baseline models, indicating the strongest probability performance. The Stacking Ensemble also got very good with a Log Loss of 0.260960, while the Random Forest achieved a value of 0.291938. On the contrary, Linear SVM and Logistic Regression produced the highest Log Loss values of 0.365625 and 0.365950, respectively. These results suggested that, although the models achieved comparable classification performance in some metrics, their ability to produce well-calibrated probability estimates was poor.

The Log Loss analysis, therefore, provided an additional perspective for comparing the baseline models and was considered together with Accuracy, Precision, Recall, F1-score, and ROC-AUC during hyperparameter tuning and final model selection.

A. Hyperparameter Tuning

Hyperparameter tuning was performed to get the best model configurations and improve the performance of the baseline models. Grid Search with cross-validation was used to evaluate predefined combinations of hyperparameters using the balanced training dataset.

For all models, multiple hyperparameter combinations were evaluated through CV. The parameter with the highest CV F1-score was selected as the optimal parameter for that model. It was configured in the validation set to determine whether hyperparameter optimization improved their classification performance.

Table VI presents the selected hyperparameter and the corresponding best cross-validation F1-scores.

TABLE VI
HYPERPARAMETER TUNING RESULTS

Model	Best Hyperparameter	CV F1-score
Logistic Regression	$C = 0.1$, solver = <code>lbfgs</code>	0.8638
Linear SVM	$C = 0.01$	0.8615
Decision Tree	max_depth = 10, min_samples_leaf = 2, min_samples_split = 5	0.9123
Random Forest	n_estimators = 100, max_depth = 20, min_samples_leaf = 2, min_samples_split = 5	0.9131
Neural Network	hidden_layer_sizes = (64, 32), learning_rate_init = 0.001	0.9119

Among the tuned models, the Random Forest achieved the highest cv F1-score of 0.9131, followed by the Decision Tree with a F1-score of 0.9123 and the Neural Network with a F1-score of 0.9119. The Logistic Regression and Linear SVM models achieved cv F1-scores of 0.8638 and 0.8615, respectively.

The tuned models were all evaluated on the validation set to determine whether the optimized hyperparameter improved their classification performance. The resulting optimized models were then used as base components for the final Stacking Ensemble, allowing the ensemble model to combine the best classification results.

B. Final Tuned Stacking Ensemble

The final model was developed using a Stacking Ensemble that combined the tuned base classifiers to leverage their complementary prediction capabilities. The tuned Logistic Regression, Random Forest, and Neural Network models were selected as the base learners, while Logistic Regression was used as the meta-learner.

The ensemble used a probability-based stacking approach using `predict_proba`, where the predicted class probabilities generated by the base learners were used as inputs to the meta-learner. The Stacking Ensemble was trained using the balanced training and evaluated on the validation set before final testing.

Table VII presents the configuration of the final tuned Stacking Ensemble.

TABLE VII
CONFIGURATION OF THE FINAL TUNED STACKING ENSEMBLE

Component	Configuration
Base Learner 1	Logistic Regression: $C = 0.1$, solver = lbfgs
Base Learner 2	Random Forest: n_estimators = 100, max_depth = 20, min_samples_leaf = 2, min_samples_split = 5
Base Learner 3	Neural Network: hidden_layer_sizes = (64, 32), learning_rate_init = 0.001
Meta-Learner	Logistic Regression
Stacking Method	Probability-based (predict_proba)

The final tuned Stacking Ensemble achieved the validation results presented in Table VIII.

TABLE VIII
VALIDATION PERFORMANCE OF THE FINAL TUNED STACKING ENSEMBLE

Metric	Validation Result
Accuracy	87.68%
Precision	69.53%
Recall	95.84%
F1-score	80.59%
ROC-AUC	0.9558

The final tuned Stacking Ensemble achieved a validation Accuracy of 87.68%, an F1-score of 80.59%, and an ROC-AUC of 0.9558. These results indicated that the tuned ensemble provided strong overall classification performance and was therefore selected for evaluation on the independent test set.

C. Final Test Evaluation

The final tuned Stacking Ensemble was evaluated using the independent test set consisting of 538,599 password instances. This test set was not used during model training or hyperparameter tuning, allowing the final evaluation to assess the generalized performance of the selected model on previously unseen data.

For model evaluation, there are used Accuracy, Precision, Recall, F1-score, ROC-AUC, and Log Loss. The results of the final test were delivered in Table IX.

TABLE IX
FINAL TEST PERFORMANCE OF THE TUNED STACKING ENSEMBLE

Metric	Accuracy	Precision	Recall	F1-score	ROC-AUC	Log Loss
Final Test Result	87.78%	69.68%	95.95%	80.73%	0.9564	0.258759

The concluding model reached an overall Accuracy of 87.78%, an F1-score of 80.73%, and an ROC-AUC of 0.9564. In particular, the model achieved a Recall of 95.95% for the Strong password class, demonstrating a strong ability to correctly identify Strong passwords. The Log Loss value of 0.258759 further indicates the quality of the model's predicted probabilities, where lower values represent better probabilistic prediction performance.

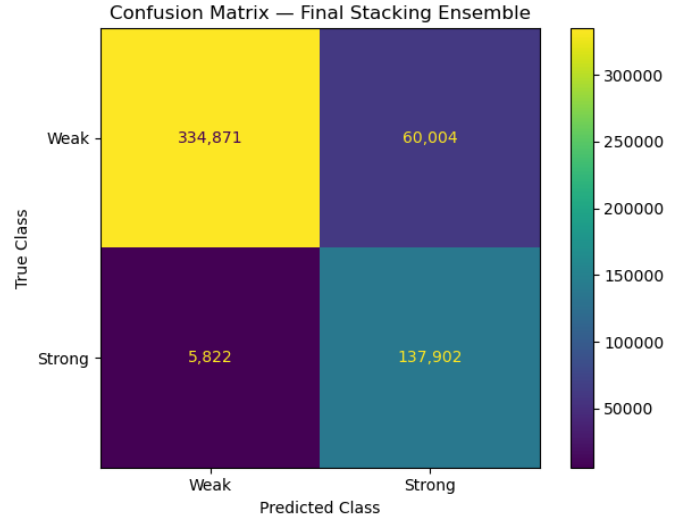


Fig. 4. Confusion matrix.

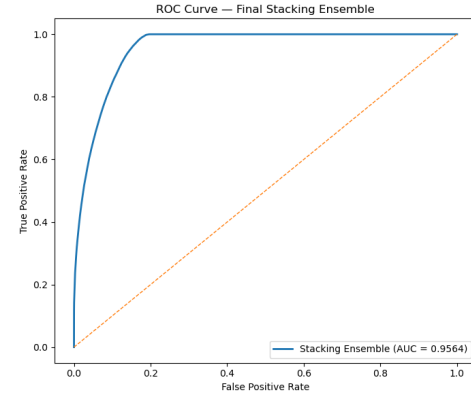


Fig. 5. ROC Curve.

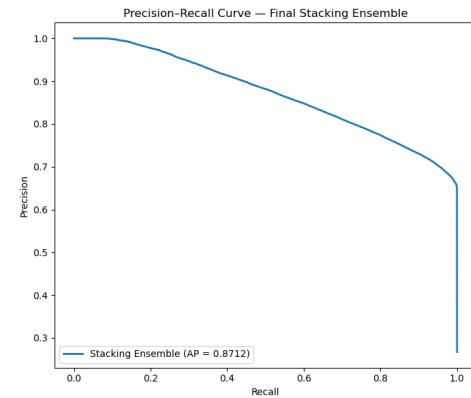


Fig. 6. Precision-Recall Curve.

The confusion matrix shows that 5,822 Strong passwords were incorrectly classified as Weak, while 60,004 Weak passwords were incorrectly classified as Strong. Overall, these results demonstrate the final performance of the proposed tuned Stacking Ensemble on previously unseen password data.

TABLE X
MCNEMAR'S TEST RESULTS

Statistical Measure	Result
Comparison	Stacking Ensemble vs. Random Forest
Test Set	538,599 samples
Contingency Table	[[464648, 3818], [8125, 62008]]
McNemar Statistic	1552.5108
p -value	< 0.001

D. Final Test Log Loss

The Log Loss of the final tuned Stacking Ensemble was calculated using the independent test set to further evaluate the quality of its predicted class probabilities. Unlike classification metrics that consider only the final predicted class, Log Loss also accounts for the confidence associated with each prediction. A lower Log Loss value indicates better probabilistic prediction performance. The final Log Loss value of 0.258759 indicates that the selected Stacking Ensemble produced strong probabilistic estimates on the previously unseen test data. This result complements the classification metrics reported in the final test evaluation and provides an additional assessment of the model's probability prediction performance.

E. Statistical Comparison

A McNemar's test was performed to statistically compare the final Stacking Ensemble with the Random Forest model using predictions generated from the same independent test set. This statistical test evaluates whether two classifiers exhibit a statistically significant difference in their classification errors when evaluated on the same observations.

The findings of the statistical analysis are shown in Table X.

The obtained p -value of less than 0.001 shows a statistically meaningful distinction in the classification behavior of the Stacking Ensemble and Random Forest models. Therefore, the analysis provides statistical evidence that the two models did not produce equivalent error patterns when evaluated on the same test observations. The McNemar's test was conducted using the non-exact formulation with continuity correction.

F. Model, Error, and Interpretability Analysis

Further analysis was performed to examine classification errors, class-wise behavior, feature importance, and computational characteristics of the evaluated models. The final Stacking Ensemble incorrectly classified 5,822 Strong passwords as Weak and 60,004 Weak passwords as Strong. Class-wise evaluation showed that the model achieved a Recall of 95.95% for Strong passwords and 84.80% for Weak passwords. These results indicate that the model was more effective at identifying Strong passwords than Weak passwords in terms of recall.

To explore the interpretability of models, the feature importance from Random Forest and the permutation feature importance were examined. Both methods recognized password length as the leading predictive characteristic. The Random Forest given a significance value of 0.693764 to password

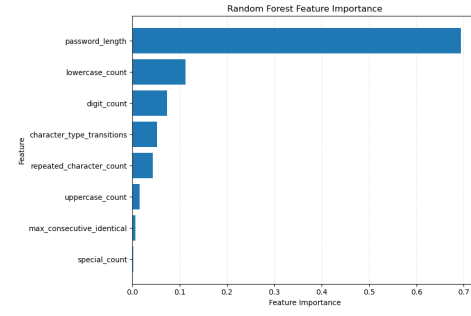


Fig. 7. Random Forest Feature Importance.

TABLE XI
MODEL ERROR AND INTERPRETABILITY ANALYSIS

Analysis	Key Result
Strong $\rightarrow$ Weak errors	5,822
Weak $\rightarrow$ Strong errors	60,004
Strong Recall	95.95%
Weak Recall	84.80%
Most Important Feature	Password Length
Random Forest Importance of Length	0.693764
Permutation Importance of Length	0.473877

length, while permutation importance produced a mean importance of 0.473877, which was substantially higher than the importance values of the remaining features.

The main findings of the error and interpretability analysis are summarized in Table XI.

The analysis performed using computational methods also showed a trade-off between predictive performance and inference speed. The Stacking Ensemble required 0.7767 seconds to process 538,599 test samples, corresponding to roughly 693,471 predictions per second, whereas the linear models achieved substantially faster inference.

Overall, the analysis signifies that the final model was particularly effective at recognizing strong passwords. Although the length of a password appeared to be the most significant predictive feature. However, the improved predictive performance of the Stacking Ensemble involves a trade-off in computational complexity and inference speed compared with simpler linear models.

VI. DISCUSSION AND LIMITATIONS

This study examined the effectiveness of multiple machine learning models for password strength classification using structural password characteristics. The results are contrasted with earlier research to identify similarities, differences, and the implications of the adopted methodology.

The present study is closely related to Password Strength Detection via Machine Learning[3], which evaluated six models: SVM, Logistic Regression, Neural Networks, Decision Trees, Random Forests, and Stacked Models. Similarly, this study evaluates Logistic Regression, SVM, Decision Tree, Random Forest, Neural Network, and Stacking models using eight structural features. The previous study reported

extremely high performance, with Decision Tree and Stacking achieving approximately 99.98% accuracy. Both studies demonstrate that machine learning models can effectively classify password strength using measurable structural characteristics.

The present study also differs in terms of dataset composition. It combines two publicly available Kaggle datasets[2][1]. After cleaning and deduplication, the final dataset contains 3,590,654 unique passwords. This provides a large collection of passwords from multiple public sources. However, a large dataset does not necessarily guarantee real-world generalizability because publicly available and leaked password datasets may represent specific user populations and password behaviors.

The present work can also be compared with research [4] that which evaluated Random Forest, SVM, CNN, and Logistic Regression using more than 660,000 passwords. That study used a broader feature framework, including password length, entropy, character diversity, dictionary matching, pattern detection, and character-level information. In contrast, the present study focuses on eight structural features: password length, digit count, lowercase count, uppercase count, special-character count, repeated characters, maximum consecutive characters, and character-type transitions. This feature set is simpler and more interpretable, but it does not capture dictionary words, semantic information, or other contextual patterns.

The results indicate that structural password characteristics can effectively distinguish between weak and strong passwords within the established labeling system. This observation aligns with previous research shows that nonlinear and ensemble models, such as Decision Trees, Random Forests, Neural Networks, and Stacking, can capture interactions between password characteristics better than simple linear models. However, high classification performance should be interpreted carefully. Since the labels in this study are derived using zxcvbn and converted into binary Weak and Strong categories, the reported Accuracy, Precision, Recall, and F1-score primarily indicate agreement with the utilized labeling system instead of concrete proof of actual password vulnerability in the real world.

The limitations of structural features are also supported by our refereed researcher[4], which showed that apparently complex passwords may still be weak because of dictionary words or predictable patterns. Since the present study does not include dictionary matching or pattern-detection features, some contextual weaknesses may not be identified.

Similarly, found that existing password strength assessment methods may struggle with non-English and multilingual passphrases. This suggests that password strength can depend not only on structural properties but also on language, character encoding, and linguistic context. Therefore, the generalizability of the present structural-feature-based approach to multilingual passwords and different password populations may be limited[6].

Another important limitation is related to privacy and secu-

rity. We found similar[5] that Demonstrated that data-driven password strength meters may be vulnerable to membership inference attacks. Therefore, password strength models should not be evaluated only based on predictive performance. Privacy and potential information-leakage risks should also be considered, particularly when models are trained using large collections of real or leaked passwords.

Overall, the findings support previous research showing that machine learning models can achieve very high performance in password strength classification. The present study contributes a large cleaned and deduplicated dataset of 3,590,654 unique passwords and compares six machine learning models using a consistent set of eight structural features. However, the results should not be interpreted as proving that password strength can be fully determined using structural characteristics alone.

The main limitations of this study are the dependence on zxcvbn-derived labels, the conversion of password strength into binary Weak and Strong categories, the absence of dictionary, semantic, keyboard-pattern, and multilingual features, and the lack of direct validation through real password-cracking experiments. In addition, the publicly available datasets may not fully represent all password populations, which may limit the generalizability of the findings. Data-driven password strength models may also introduce potential privacy or information-leakage risks.

Therefore, the high performance achieved by the models should be interpreted as strong agreement with the zxcvbn-derived Weak and Strong labels rather than as direct proof of real-world password resistance or crackability. Nevertheless, the results demonstrate that structural password characteristics provide useful information for machine-learning-based password strength classification.

VII. CONCLUSION

This study investigated the effectiveness of multiple machine learning models for password strength classification using structural password characteristics. Two publicly available Kaggle datasets [2][1], were used. After data cleaning and deduplication, a dataset containing 3,590,654 unique passwords was obtained.

Eight structural features were extracted from each password: password length, digit count, lowercase count, uppercase count, special-character count, repeated character count, maximum consecutive identical characters, and character-type transitions. These features were used to train and evaluate Logistic Regression, SVM, Decision Tree, Random Forest, Neural Network, and Stacking Ensemble models for Weak and Strong password classification.

The results showed that structural password characteristics can effectively find differences between the adopted password-strength categories. Among the evaluated models, the Stacking Ensemble achieved the strongest overall performance, with 87.78% accuracy, 80.73% F1-score, 95.95% recall, and 0.9564 ROC-AUC. Additional analyses using Log Loss, statistical comparison, error analysis, feature importance, and computational efficiency provided a broader evaluation of model

performance. Password length emerged as the most influential feature within the engineered feature set.

However, the findings should be interpreted within the scope of the adopted labeling methodology implemented in this experiment. Since the target labels were derived using zxcvbn, the models generally learned to classify zxcvbn-derived Weak and Strong categories rather than directly measuring real world password vulnerability. The results demonstrated the usefulness of automated password strength classification. Future work may incorporate richer password features, multiclass classification, external validation, and evaluation against real password-cracking scenarios.

REFERENCES

- [1] Babar Ali Jamali, “Leaked passwords largest datasets,” <https://www.kaggle.com/datasets/babaralijamali/leakedpasswords>, accessed: 2026-08-26.
- [2] Programmerrdai, “Ultimate cybersecurity password & username dataset,” <https://www.kaggle.com/datasets/programmerrdai/brute-force-database>, accessed: 2026-08-26.
- [3] J. Mo, H. Kuang, and X. Li, “Password strength detection via machine learning: Analysis, modeling, and evaluation,” *arXiv preprint arXiv:2505.16439*, 2025. [Online]. Available: <https://arxiv.org/abs/2505.16439>
- [4] M. E. M. Mazelan, N. H. A. Mutalib, and N. AlDahoul, “Enhancing password security through a high-accuracy scoring framework using random forests,” *arXiv preprint arXiv:2511.09492*, 2025. [Online]. Available: <https://arxiv.org/abs/2511.09492>
- [5] M. Xu, W. Han, J. Yu, J. Liu, X. Zhang, Y. Lin, and J. S. Dong, “On the account security risks posed by password strength meters,” in *Proceedings of the 20th ACM Asia Conference on Computer and Communications Security*, 2025, pp. 1582–1599.
- [6] C.-e. A. Tai, U. Hengartner, and A. Wong, “Evaluating the strength and availability of multilingual passphrase authentication,” in *Workshop on Measurements, Attacks, and Defenses for the Web (MADWeb 2025)*, San Diego, CA, USA, 2025.